Muhammad Abdullah Goher

mgoher@seas.upenn.edu | (215) 578-1132 | LinkedIn | GitHub | abdullahgoher.com

Education

University of Pennsylvania — Aug 2023 – May 2027

B.S.E. & M.S.E. in Computer Science (Submatriculation), Concentration in Artificial Intelligence — *Philadelphia, PA*

GPA: 3.71 (B.S.E.) / 3.85 (M.S.E.) | **Coursework:** Applied Machine Learning, Artificial Intelligence, Big Data Analytics, Computer Graphics, Operating Systems, Databases, Probability, Statistical Inference

Experience

AeternalLabs, PBC – Founding AI Engineer — May 2026 – Present

Runtime AI-governance platform auditing LLMs for demographic fairness (first engineering hire) — *Remote*

- Built a multi-provider LLM routing layer spanning **8 providers and 18+ frontier models** (OpenAI, Anthropic, Google, xAI, DeepSeek, Cerebras), unifying inference behind a single FastAPI gateway.
- Parallelized counterfactual A/B model evaluation with a thread pool, cutting end-to-end runtime **1.88**× (158s → 84s).
- Architected a baseline-context caching layer to eliminate redundant demographic-swap LLM calls, targeting a ∼**10**× reduction in API spend.
- Designed a PostgreSQL audit-trail store (SQLAlchemy) with SHA-256 input-integrity hashing for tamper-evident, regulator-ready provenance.

Computational Social Science Lab (Prof. Duncan Watts) – Machine Learning Engineer — May 2025 – Dec 2025

Human-mobility research platform & large-scale geospatial analysis — *Philadelphia, PA*

- Built an LLM-powered pipeline extracting structured JSON features from **100+** scholarly papers, plus a RAG layer surfacing quantitative mobility metrics.
- Engineered the platform backend (Express, MongoDB, MVC) with secure REST APIs and AWS S3 storage; added Cypress/Vitest tests, parallelized to cut test runtime **50%**.
- Modeled crowd density from multi-terabyte GPS datasets using PySpark on AWS EMR for 2026 FIFA World Cup planning.

PennAdapt – Machine Learning Engineer — Sep 2024 – Present

Penn Assistive Devices & Prosthetic Technologies — *Philadelphia, PA*

- Built a real-time computer-vision model (CNN + Vision Transformer, PyTorch) detecting **26 classes** of surgical tools at **90%+** accuracy across varying angles.
- Curated and annotated a **1,500+** image dataset with Penn Medicine surgeons within a 10-engineer team.

Clab AI – Artificial Intelligence Intern — May 2024 – Aug 2024

AI-powered college-application assistant used by 100+ students — *Hybrid – Nashville, TN*

- Fine-tuned LLMs on hundreds of accepted essays using RLHF to personalize writing guidance.
- Built hyperparameter-tuned Random Forest / Linear Regression models predicting financial-aid eligibility at **82%** R^2, with EDA across 100+ universities.

Projects

Research Swarm | *Python, LangGraph, LangChain, FastAPI*

- Built a LangGraph multi-agent research system (converge–diverge): a planner dynamically fans out parallel research branches, then a *cross-vendor* critic verifies key claims to cut correlated hallucination.
- Orchestrated 5+ LLM providers (Claude, GPT o-series, Gemini, Perplexity, local Ollama) by task strength, with budget- and round-capped re-diverge loops feeding a cited synthesis.

News Source Classification | *PyTorch, HuggingFace Transformers, scikit-learn*

- Co-built (team of 3) a headline classifier distinguishing Fox vs. NBC; scaled the dataset **3,800** → **17,900** headlines via multithreaded sitemap and Apify scraping pipelines.
- Fine-tuned RoBERTa/DeBERTa backbones; best model (RoBERTa-base + LinearSVC head on [CLS] embeddings) reached **91.75% accuracy**, +25 pts over a TF-IDF baseline.

Technical Skills

AI / ML: PyTorch, TensorFlow/Keras, scikit-learn, HuggingFace Transformers, LLM fine-tuning (LoRA/PEFT, RLHF), RAG, LangChain, LangGraph, CNNs/ViTs, NLP, XGBoost

Languages: Python, C, C++, Java, JavaScript/TypeScript, SQL

Data: pandas, NumPy, PySpark, vector databases (FAISS/Pinecone/Chroma), Matplotlib, Jupyter

Backend & Infra: FastAPI, Node.js, Express, PostgreSQL, MongoDB, Docker, AWS (S3, EMR), Git, Linux